

Pruthvi Sanghavi

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Robotics Systems Engineer with 5+ years driving system-level architecture, requirements engineering, and cross-functional integration on commercially deployed autonomous platforms. Deep experience in V&V, HITL/SIL testing, DVP&R, functional safety (PLd, ISO 10218), and interface management across mechanical, electrical, firmware, and software domains in safety-critical, regulated environments. Proven track record translating field data and operational constraints into measurable system improvements in fast-paced startup environments.

EXPERIENCE

Miso Robotics Inc.

Pasadena, California

Principal Systems Engineer

April 2026 – Present

- Defined and owned the systems engineering roadmap, establishing org-wide processes for software requirements management, V&V planning, and release readiness aligned to the SDLC V-model.
- Represented systems engineering in executive and stakeholder reviews, translating complex technical tradeoffs into actionable product decisions and aligning cross-functional teams on priorities.

Lead Systems Engineer

January 2025 – March 2026

- Led EVT, DVT, and PVT phases for Flippy 3, coordinating cross-functional readiness gates across mechanical, electrical, firmware, and software teams to drive hardware validation from prototype to production release.
- Authored and managed 200+ subsystem requirements and 15+ ICDs across mechanical, electrical, firmware, and software interfaces, maintaining end-to-end verification traceability matrices with 95% requirement coverage.
- Analyzed field performance data from 15+ production robots to proactively identify failure modes, driving an automated fault recovery initiative that raised fleet NPS from 5/10 to 8/10 within one week of deployment.
- Built automated test infrastructure using ROSTests, PyTests, and Playwright covering smoke, stress, performance, and fault injection testing - reducing testing time by 40% and regression turnaround to 2 days.
- Defined and monitored 7+ KPIs tracking fleet reliability and operational efficiency, translating deployment data into actionable design improvements and go/no-go release decisions for leadership.

Systems Engineer

July 2021 – December 2024

- Architected and maintained the functional safety system for Flippy 3, achieving Performance Level D (PLd) through comprehensive hazard analysis, risk assessment, and ISO 10218 compliance validation.
- Designed and built HITL test fixtures and validation infrastructure for subsystem-level testing in ROS, enabling accelerated life cycle testing across millions of cycles and surfacing critical failure modes ahead of vehicle integration.
- Led complete hardware/software integration of Yaskawa GP4 manipulator including controller configuration, ROS driver development, and cross-functional coordination with the vendor engineering team.
- Performed in-depth root cause analyses (RCAs) and applied CAPA using flowcharts and fishbone diagrams, reducing fleet downtime by 20% and improving overall system reliability.
- Managed ML models for a food classifier spanning 100+ classes using GCP VertexAI (95% accuracy, 99% precision) and implemented a CI/CD pipeline for continuous model retraining and deployment.

Rezoom Lab, University of Maryland

College Park, Maryland

Graduate Robotics Researcher / Entrepreneurial Lead

January 2020 – May 2021

- Developed an autonomous ROS software stack for an E-Scooter including component specification, software interface definition, digital twin modeling, and SIL testing for sensor/actuator validation.
- Implemented multi-sensor fusion using Extended Kalman Filtering for autonomous navigation, achieving sub-meter positioning accuracy by integrating GPS, Visual-Inertial Odometry, and wheel encoder data streams.
- Led the NSF I-Corps project from proposal to stakeholder engagement, conducting 120+ customer interviews and securing \$50,000+ in grants through investor pitches and industry expert validation.

EDUCATION

University of Maryland, College Park

Masters of Engineering, Robotics (2021)

LDRP Institute of Technology and Research

Bachelor of Engineering, Mechanical Engineering (2019)

SKILLS

Core Programming: Python, C++, Matlab, SQL

Frameworks/Libraries: ROS, MuJoCo, OpenCV, Grafana, Linux, Docker, Git, Jira, GCP, PyTest, Playwright

SW Methodologies: Agile, Scrum, V-Model, DVP, OOP & Procedural Programming, CI/CD, TDD

Safety Standards: ISO 10218, Hazard Analysis, Risk Assessment

Hardware Platforms: Yaskawa and Fanuc Manipulators, Teknik and Dynamixel Motor Stack, Keyence & Sick Safety PLCs

Communication Protocols: Modbus, FTP, RS-232/UART, RTSP, TCP/IP

Diagnostics Tools: Oscilloscope, Logic Analyzer, Wireshark, Multimeter

Design/Modeling Tools: Solidworks, Onshape, Blender